# Edema :

Edema consed by excissive accumulation of fluid within the interstitial space.

## Causes of edema :

1. increse total extracellular fluid
2. High local venous Presure
3. low plasma oncotic Presure / Serum albumin
4. incresed capillary Permabillity
5. lymphatic obstruction

**pitting edema :**
indentation appears and Persists when Presure is applied.

**non pitting edema :**
indentation doesnt form, is typical of lymphatic obstruction, may also result of excessive matrix deposition in tissues, ex/ hypothyrodism

## investigation :

1. examination of the cardiovascular System and abdomen
2. blood tests (electrolytes, Serum albumin, urine test for Protein)
3. imaging of the organs like (liver, Kidney, heart)
4. where acsites and plural effusion accur aspiration of fluid with measurment of proteins and glucose, microscopy for cells → for distinguishing between transudate (generalized edema) and exudate (localized edema)

# Cyanosis :-

Bluish color of the skin, resulting from an increased quantity of reduced hemoglobin (deoxgenated hemoglobin)

## Causes of central cyanosis :-

1. decresed arterial oxgen saturation
2. decresed atmospheric presure
3. impaired pulmunary function
4. alveolar hypoventilation
5. inhomogenity in pulmonary ventilation
6. impaired oxgen diffusion
7. Hemoglobin with low affinity for oxgen
8. Hemoglobin abnormality
9. Congenital heart disease

decreased $SaO_2$
↓
results from reduction of
↓
$PaO_2$
↓
brought about by a decline in
↓
$FiO_2$

## Causes of Peripheral cyanosis :-

1. reduced cardiac output
2. Cold exposure
3. Redistribution of blood flow from extrimities
4. Arterial obstruction
5. venous obstruction

the most common cause of Peripheral cyanosis is normal vasoconstriction

Certain features are important in arriving at the cause of cyanosis:

1. It is important to ascertain the time of onset of cyanosis.
2. Central and peripheral cyanosis must be differentiated
3. The presence or absence of clubbing of the digits should be ascertained
4. Pao2 and Spo2 should be determined